



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

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Experiment No. 6

Title: Use string functions and regex for email validation, data extraction, and text analysis.

Software/Tools Required: HTML5, CSS3, JavaScript, a web browser such as Google Chrome or Microsoft Edge, and a code editor such as Visual Studio Code.

Theory

1. String Functions

JavaScript string functions are used to process and analyze textual data. Operations such as `toLowerCase()`, `match()`, `replace()`, and `length` help transform text, identify content, count characters, and prepare information for further processing. In the case study, string processing is applied to the complete paragraph entered by the user.

2. Regular Expressions

Regular expressions, or Regex, are patterns used to search, identify, extract, and validate specific forms of text. JavaScript supports Regex through methods such as `match()` for extracting matching content and `test()` for checking whether text follows a required pattern. This makes Regex suitable for locating structured information inside paragraph-form input.

3. Email Validation

Email validation checks whether an extracted address follows the expected email structure. The program first extracts the email using Regex and then validates it with a separate expression. The result is displayed as Valid or Invalid.

4. Data Extraction

Data extraction identifies required information from unstructured text. The implemented program uses Regex patterns to extract the student's name, email address, ten-digit phone number, and department. If a value is not found, the interface displays 'Not found', allowing the user to clearly identify missing information.

5. Text Analysis

Text analysis provides basic numerical and transformation-based information about the input. The program calculates the total number of words and the total number of characters excluding spaces. It also converts the processed paragraph to lowercase and replaces occurrences of Computer Science with Information Technology before displaying the processed text.

Case Study Implemented

The case study is a Student Information Extractor developed using `index.html`, `script.js`, and `style.css`. The HTML page accepts student information in paragraph format and displays the name, email and status, phone and status, department, word count, character count, and processed text. The JavaScript uses Regex patterns to extract and validate the name, email, phone, and department. It also calculates word and character counts, replaces Computer Science with Information Technology, and updates the results through DOM manipulation. The CSS provides the card layout, input area, button, result section, validation styles, and responsive design.

Conclusion

The case study successfully demonstrates how string functions and regular expressions can be used for email validation, data extraction, and text analysis. The implementation provides a simple web-based solution for processing paragraph-form student information while displaying validated and analyzed results.